F1 second pass: partial source and derived extraction design

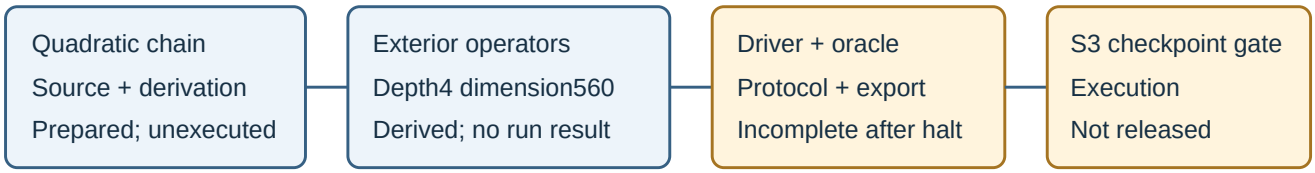

Figure 1: Editable study diagram; exact scope and evidence paths appear in the report.

26 October 2026. This is a Phase A preparation report. No second-pass mathematical program was imported or executed, and no result on the fixed depth-four cases is claimed.

The implementation worker prepared core.py, exterior.py, inputs.json, derivation.md and literature.md before its turn was halted by a platform content flag. Their exact bytes are recorded in partial-source-manifest.json. The failure and the coordinator's subsequent model-capacity error are retained in ../../environment/agent-halts-01.json. These are operational events, not evidence about whether the proposed mathematics works.

Prepared design

The proposed method represents the scalar Dickson fibre by its quadratic-chain quotient, constructs multiplication operators on the exterior cube, evaluates the fixed symmetric S4 relation and extracts complete rational characters from its exact kernel. The distinct-state representation at depth4 has dimension C(16,3)=560. Two-equal and all-equal strata use separate tensor representations. These are derived representation sizes and arguments, not observed kernel dimensions or measured runtimes.

derivation.md states the assumptions, the no-root-enumeration boundary of the primary extractor, the separate enumerative arity-four control and the explicit matrix growth. literature.md records actual source reading and unresolved solving complexity. The previous depth-four watchdog results remain unknown; source preparation does not change them.

What is missing

The retained source has not been checked through a complete driver, independent point-domain oracle, fixed-panel analysis/export and validated prospective protocol. Those interfaces must be completed and statically inspected before a freeze. The arity-four positive-control construction also needs an explicit availability/refusal rule for rational lifts rather than an assumption that every algebraic coordinate partition has a rational positive in the chosen fibre.

No frozen protocol, release, experimental row or passing mathematical certificate is supplied for this partial package. The S3 disclosure approval and actual checkpoint gate remain separate execution requirements. Root preserves the partial implementation and finishes unaffected custody/reporting work; the blocked sub-agent request is not retried or rephrased.

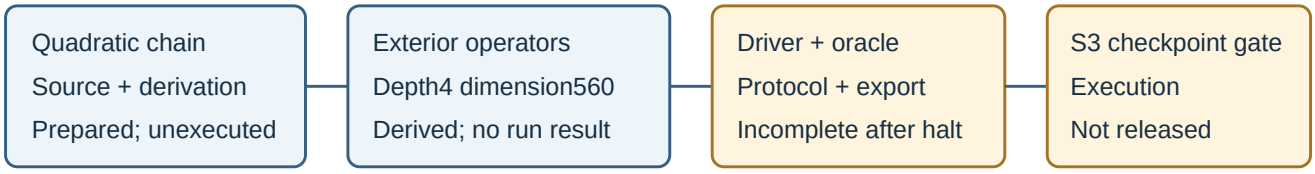
Quadratic chain
Source + derivation
Prepared; unexecuted
Exterior operators
Depth4 dimension560
Derived; no run result
Driver + oracle
Protocol + export
Incomplete after halt
S3 checkpoint gate
Execution
Not released
TASK-20261026-a0cc5b: operational halt is preserved; no fixed-case or complexity conclusion follows.

Figure 2: F1 preparation and remaining interfaces